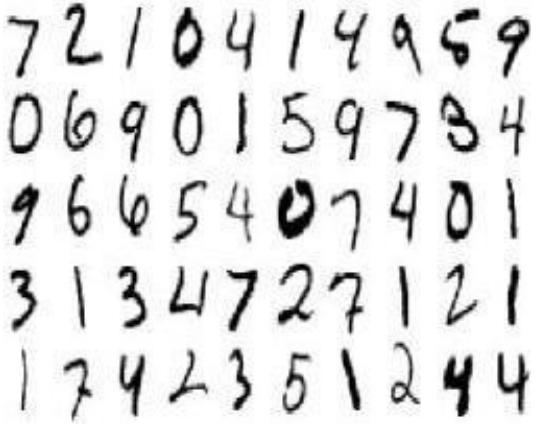
K-Nearest Neighbour Classifier

Acknowledgements: Lecture slides material from Duda, Hart and Stork, Dr. Gavin Brown
Pattern Recognition and Analysis: Dr Arsalan Shaukat

Problem Statement

- Why recognising rugby players is (almost) the same problem as *handwriting recognition*



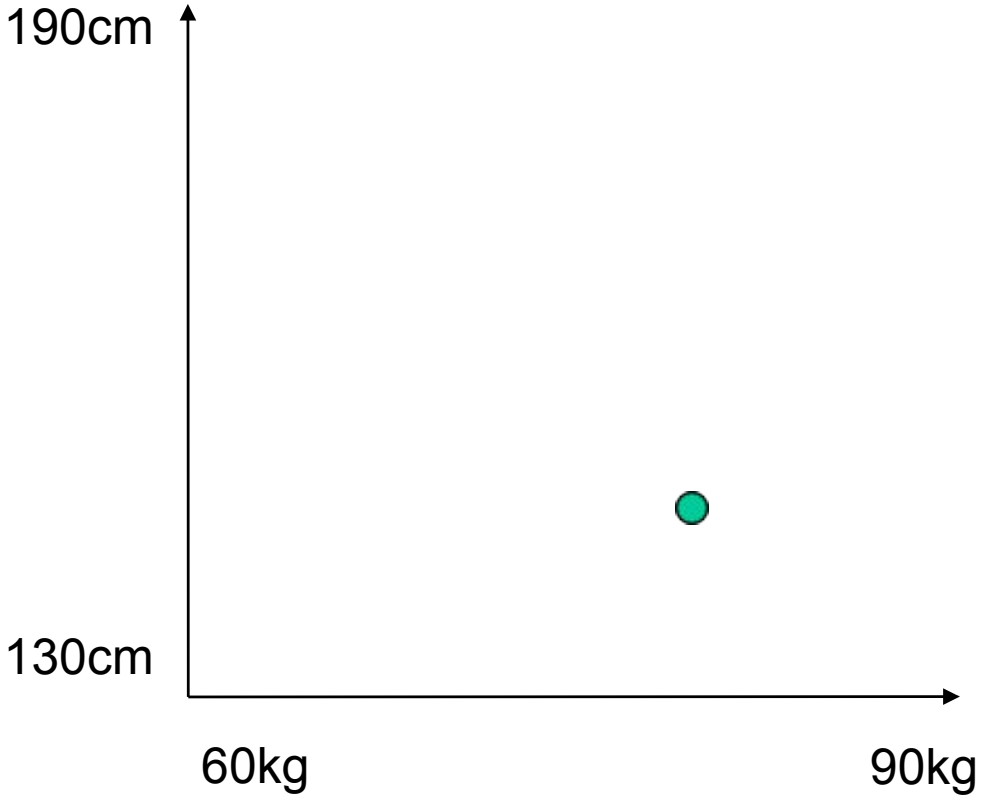
Problem Statement

Can we LEARN to recognise a rugby player and ballet dancer?

What are the “features” of a rugby player?

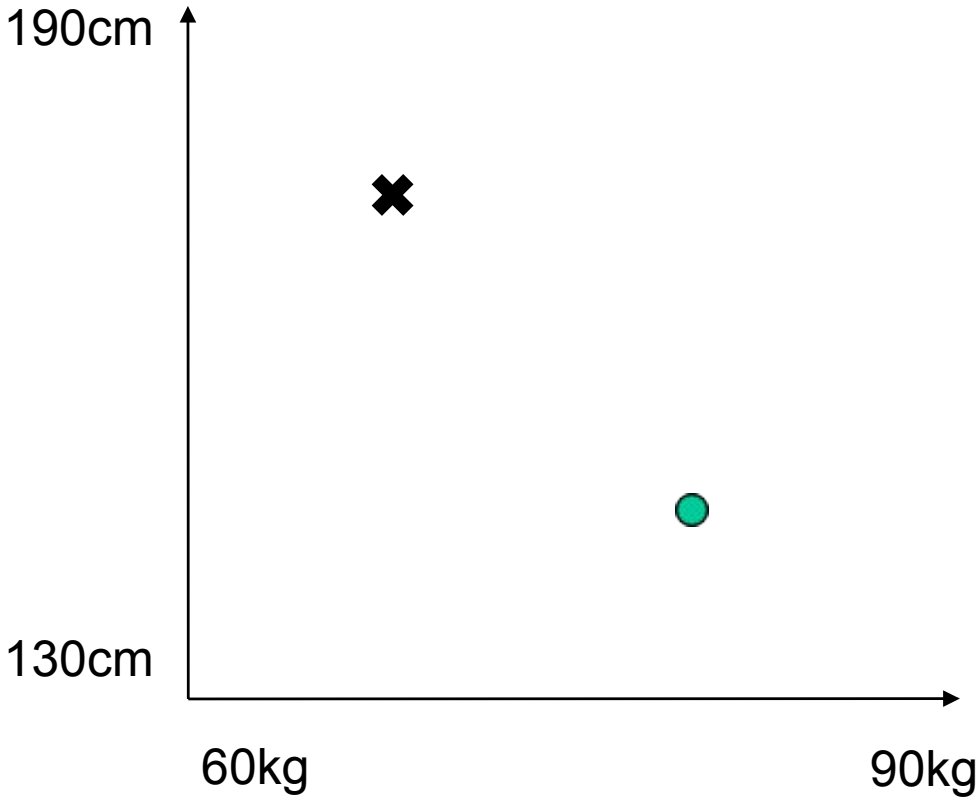
Features

Rugby players = short + heavy?



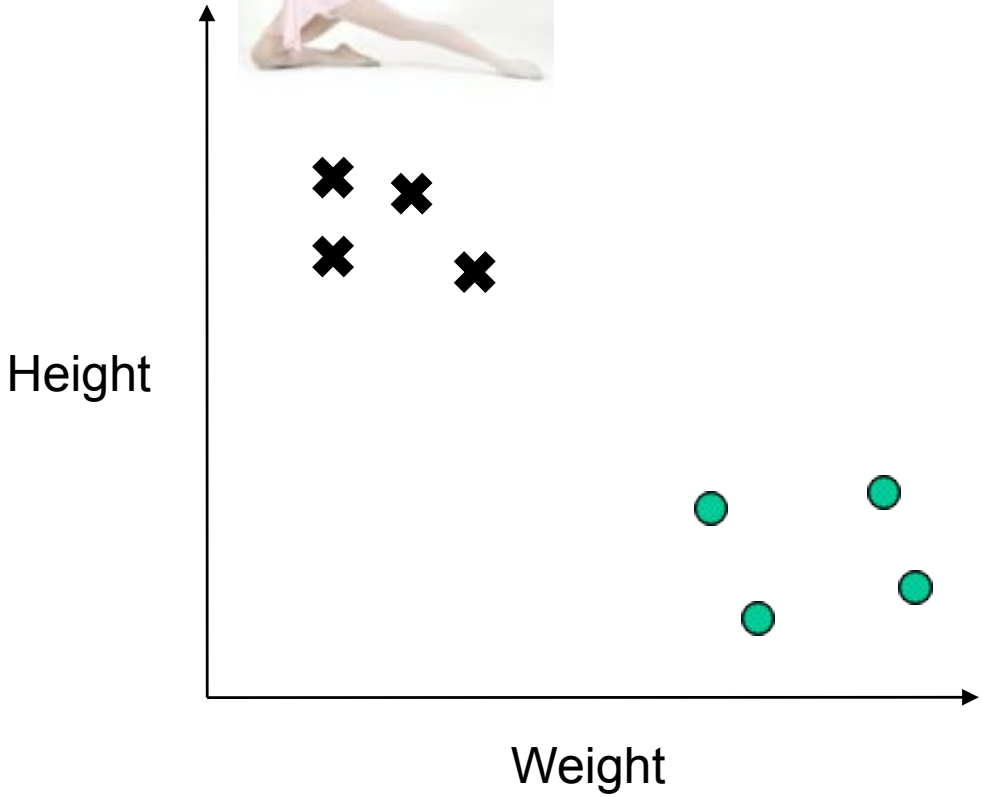
Features

Ballet dancers = tall + skinny?

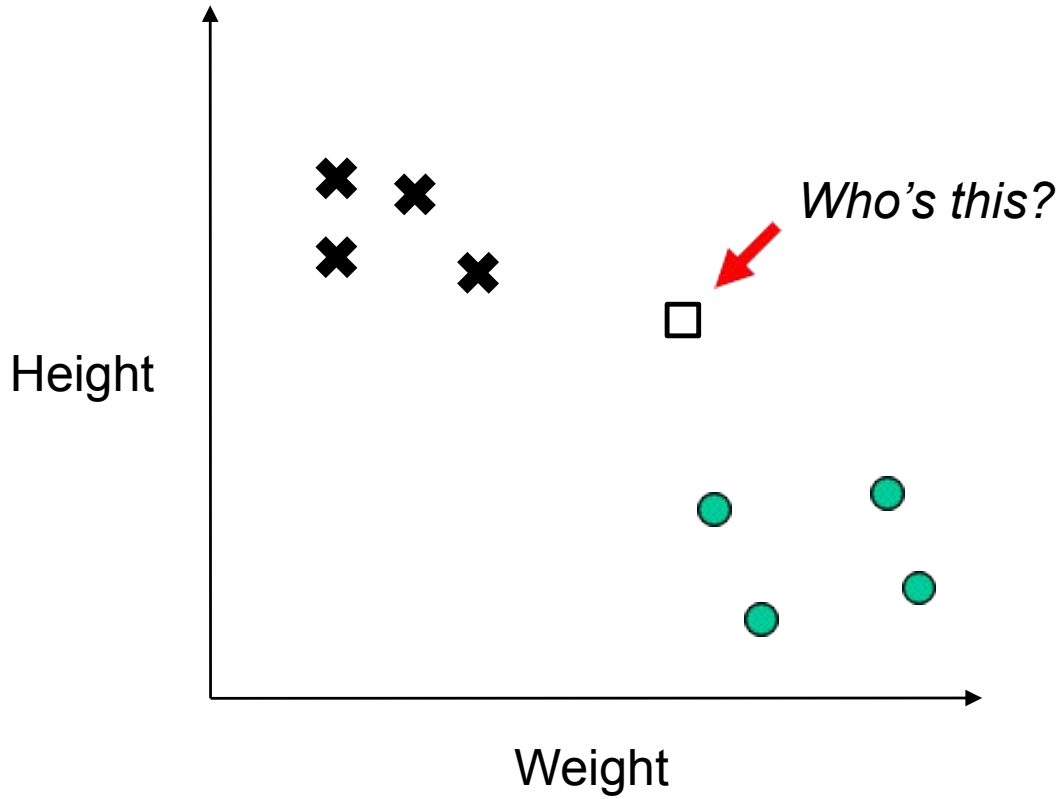


Feature Space

Rugby players “cluster” separately in the space.

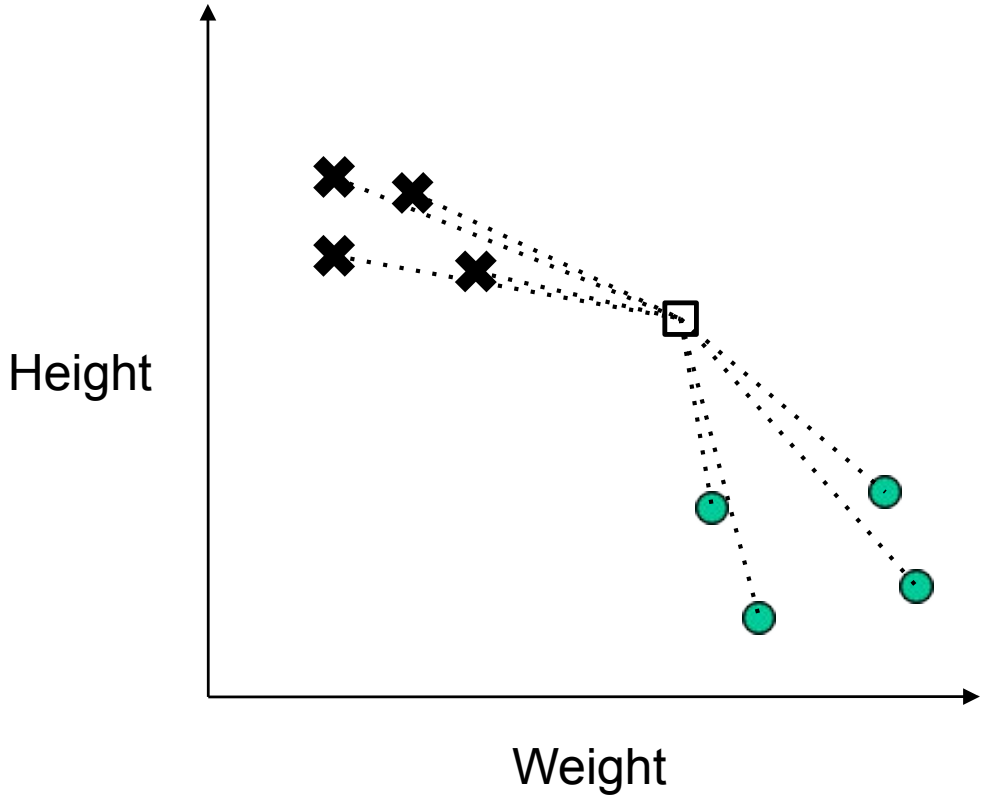


The K-Nearest Neighbour Algorithm



The K-Nearest Neighbour Algorithm

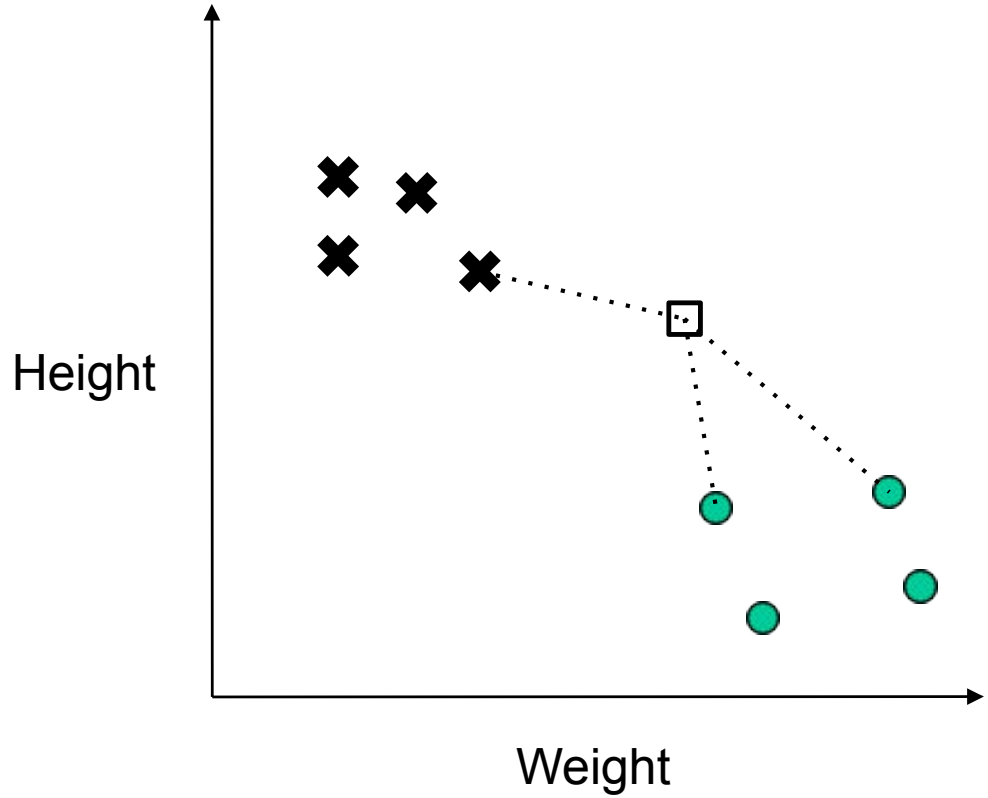
1. Measure distance to all points



The K-Nearest Neighbour Algorithm

1. *Measure distance to all points*
2. *Find closest "k" points*

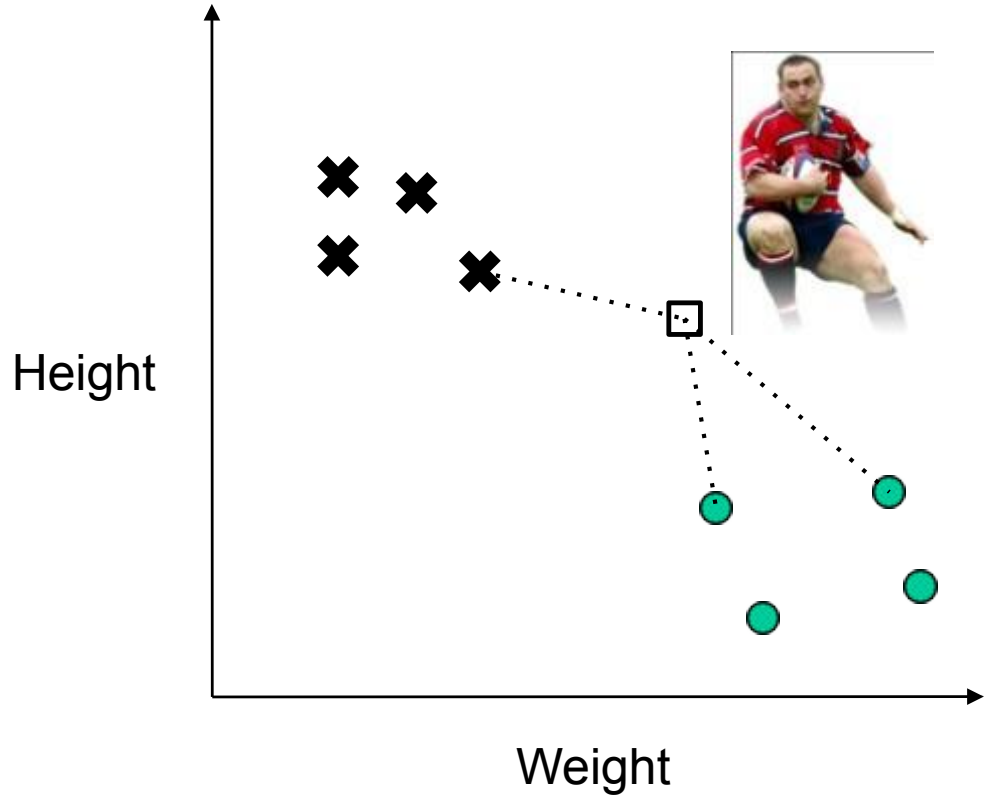
(here k=3, but it could be more)



The K-Nearest Neighbour Algorithm

1. *Measure distance to all points*
2. *Find closest "k" points*
3. *Assign majority class*

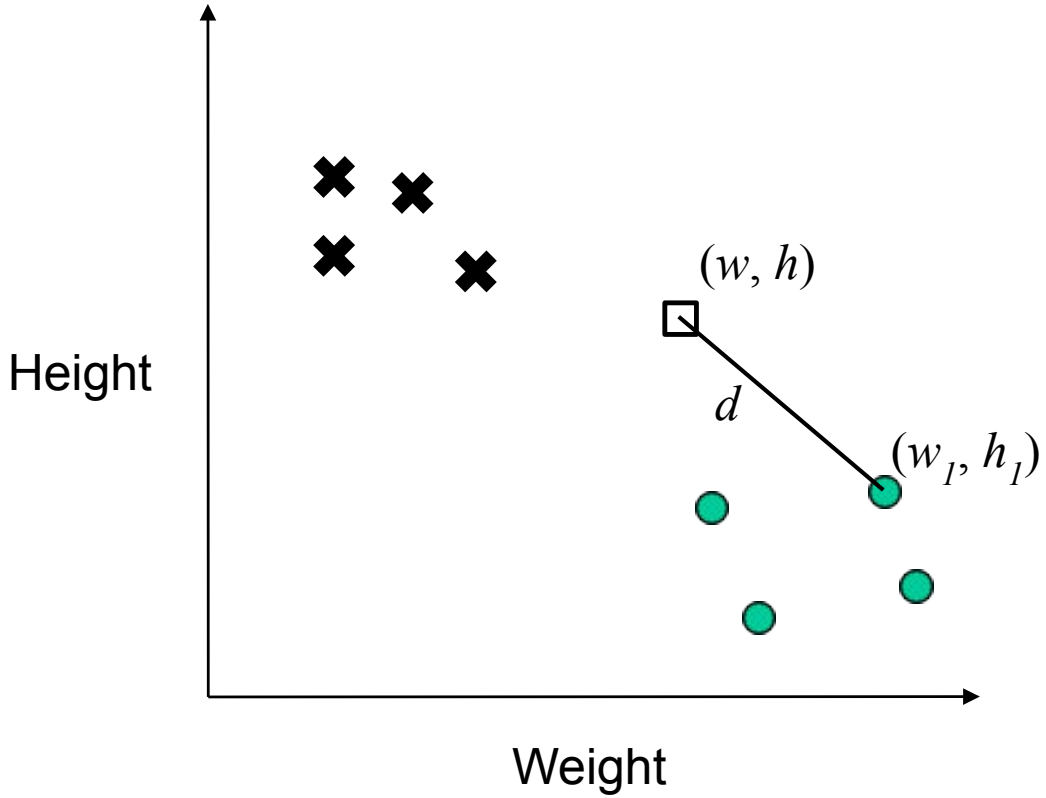
 (here k=3, but it could be more)



Distance Measure

"Euclidean distance"

$$d = \sqrt{(w - w_1)^2 + (h - h_1)^2}$$



The K-Nearest Neighbour Algorithm

for each testing point

measure distance to every training point

find the k closest points

identify the most common class among those k

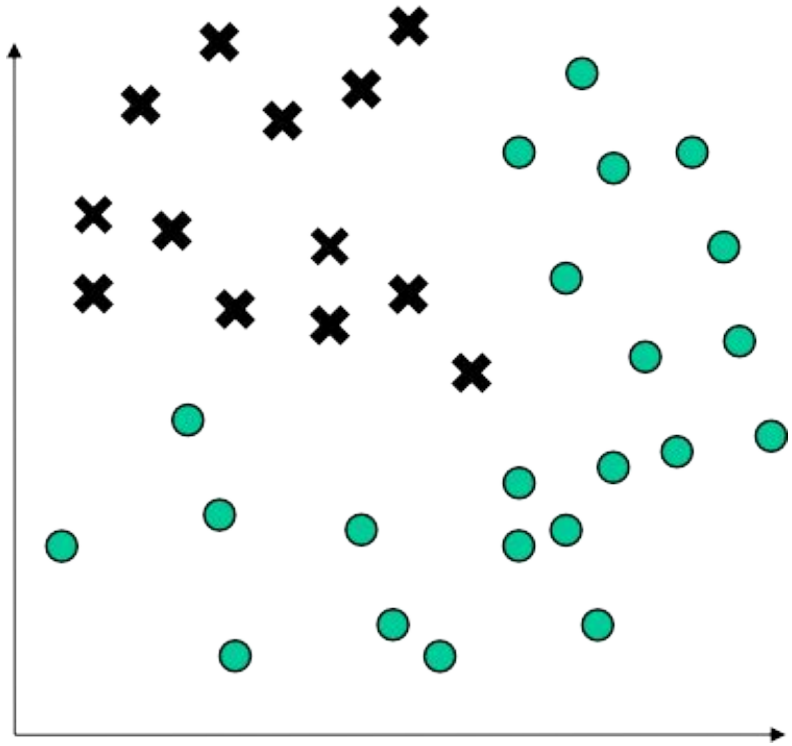
assign that class

end

- Advantage: Surprisingly good classifier!
- Disadvantage: Have to store the entire training set in memory

The Decision Boundary

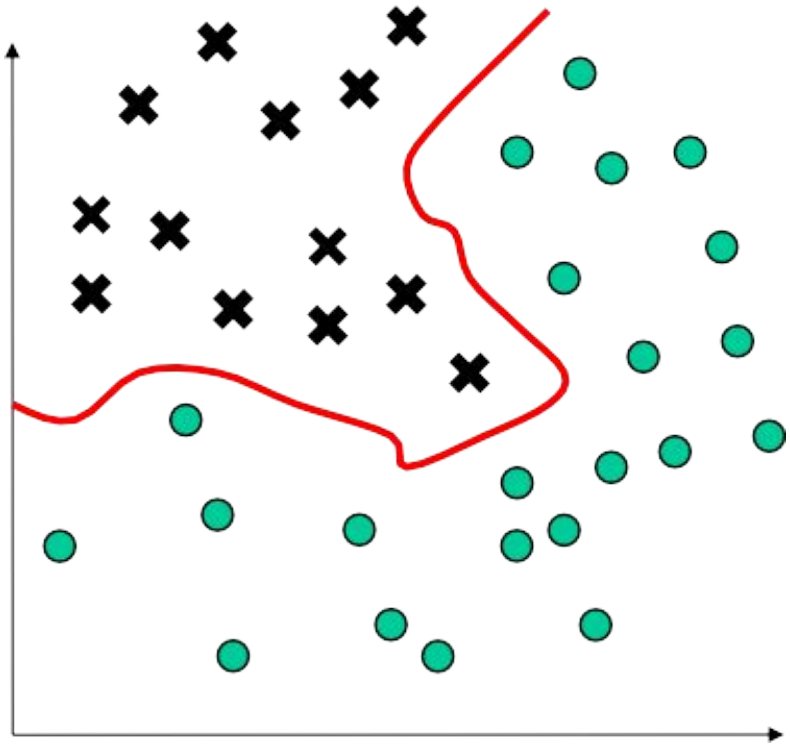
Where's the decision boundary?



The Decision Boundary

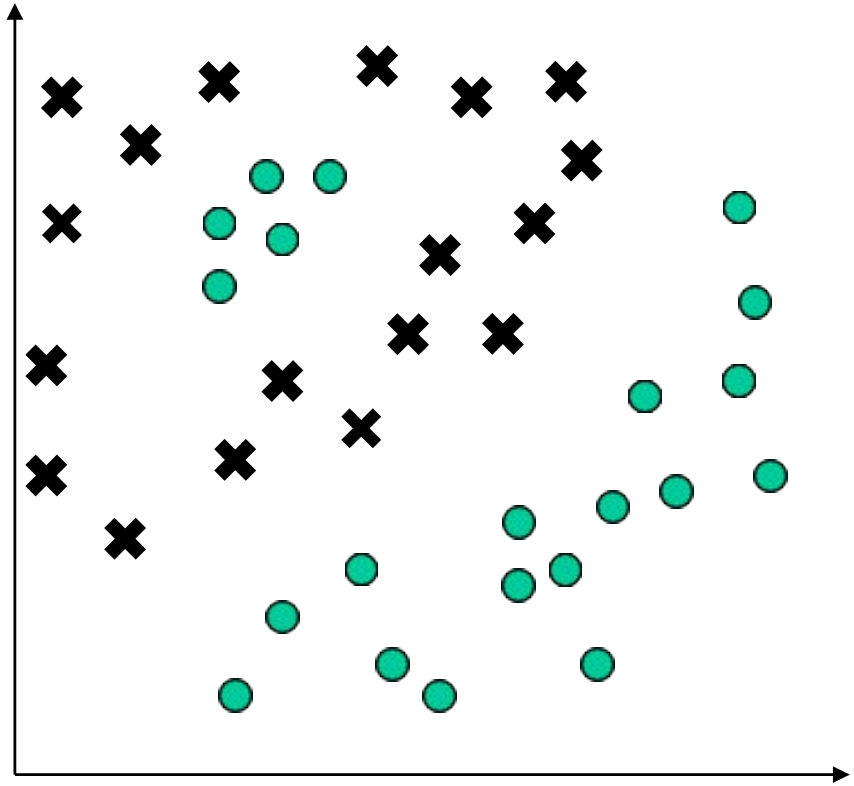
Where's the decision boundary?

Not always a nice neat line!



The Decision Boundary

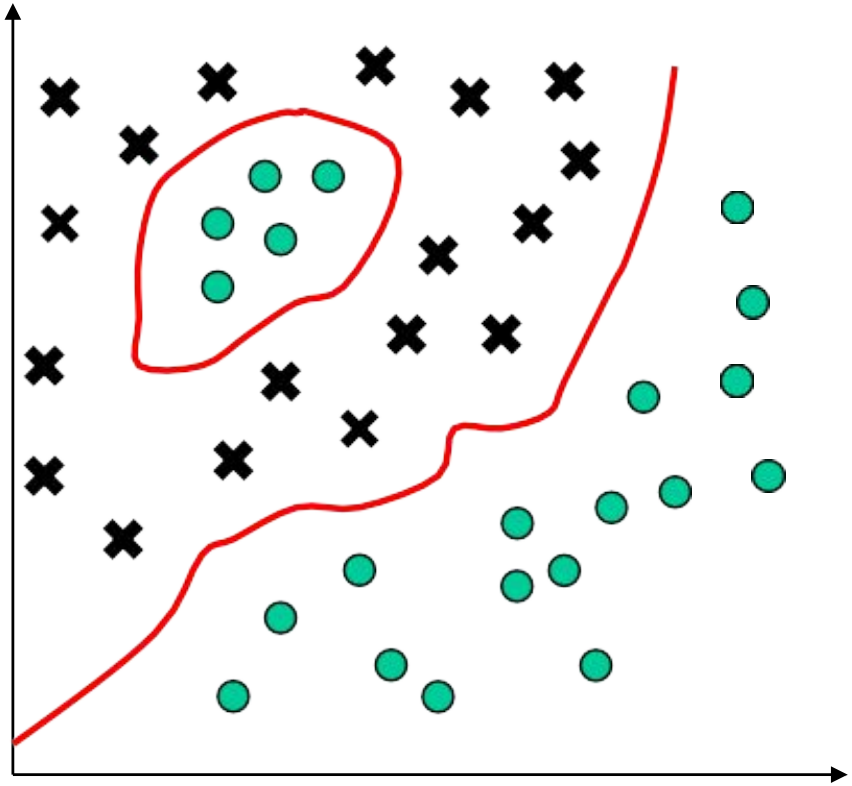
Where's the decision boundary?



The Decision Boundary

Where's the decision boundary?

Not always contiguous!



Distance Measure

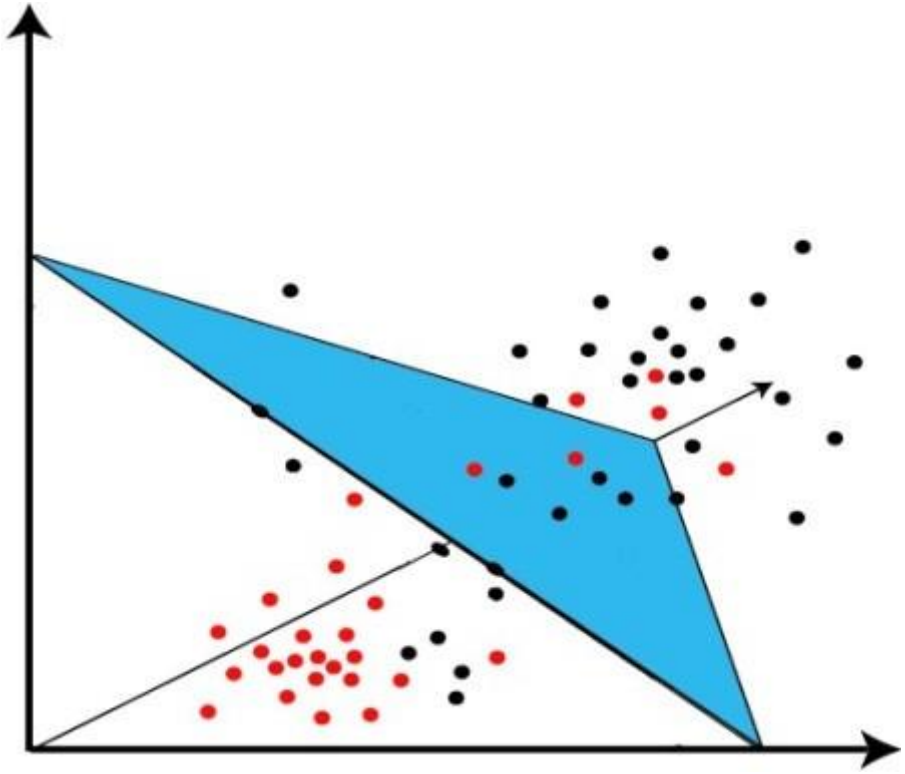
Euclidean distance still works in 3-d, 4-d, 5-d, etc….

$$d = \sqrt{(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2}$$

x = Height
y = Weight
z = Shoe size

The Decision Boundary

Called a decision **surface** in 3-d and upward…

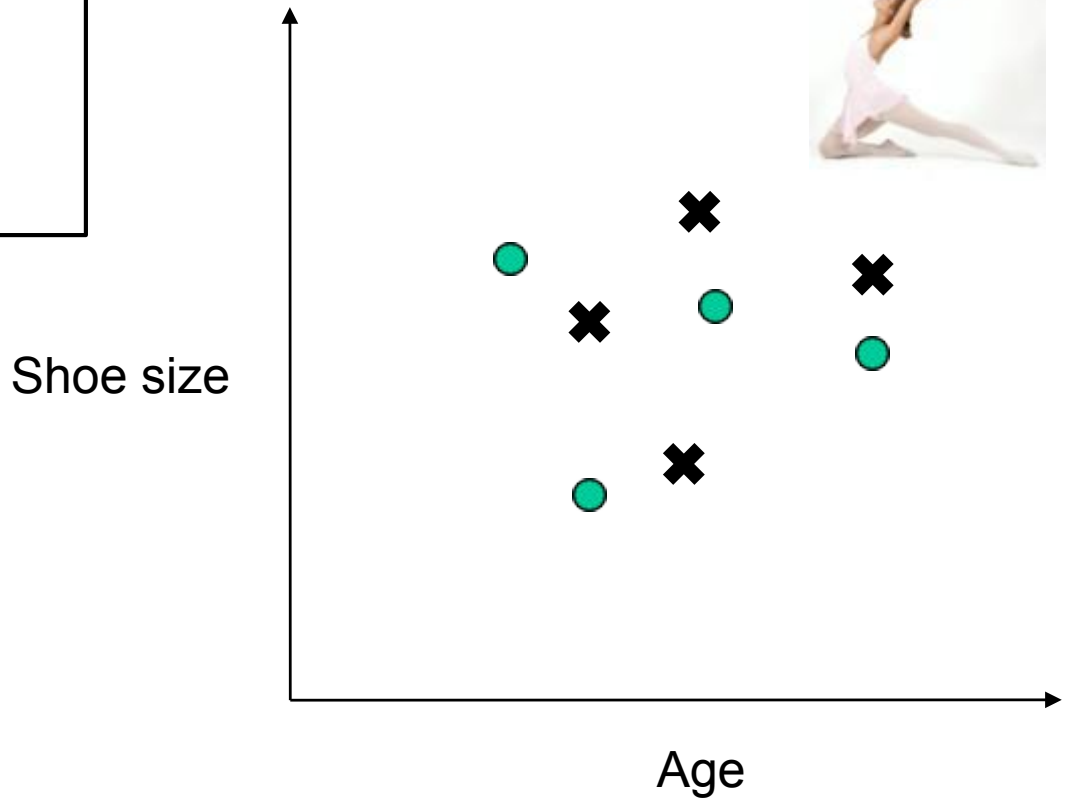


Features

Choosing the wrong features makes it difficult
Too many features
It's computationally intensive

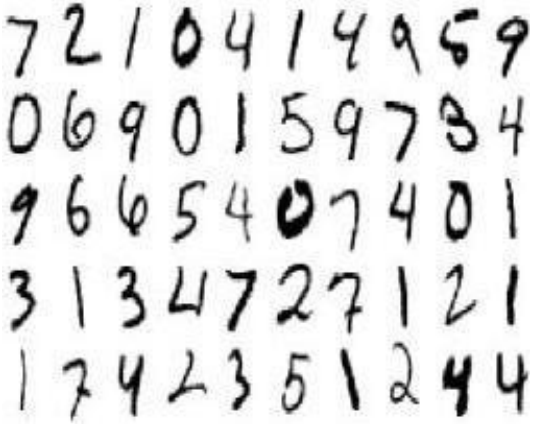
Possible features:
- Shoe size ✓
- Height
- Age ✓
- Weight

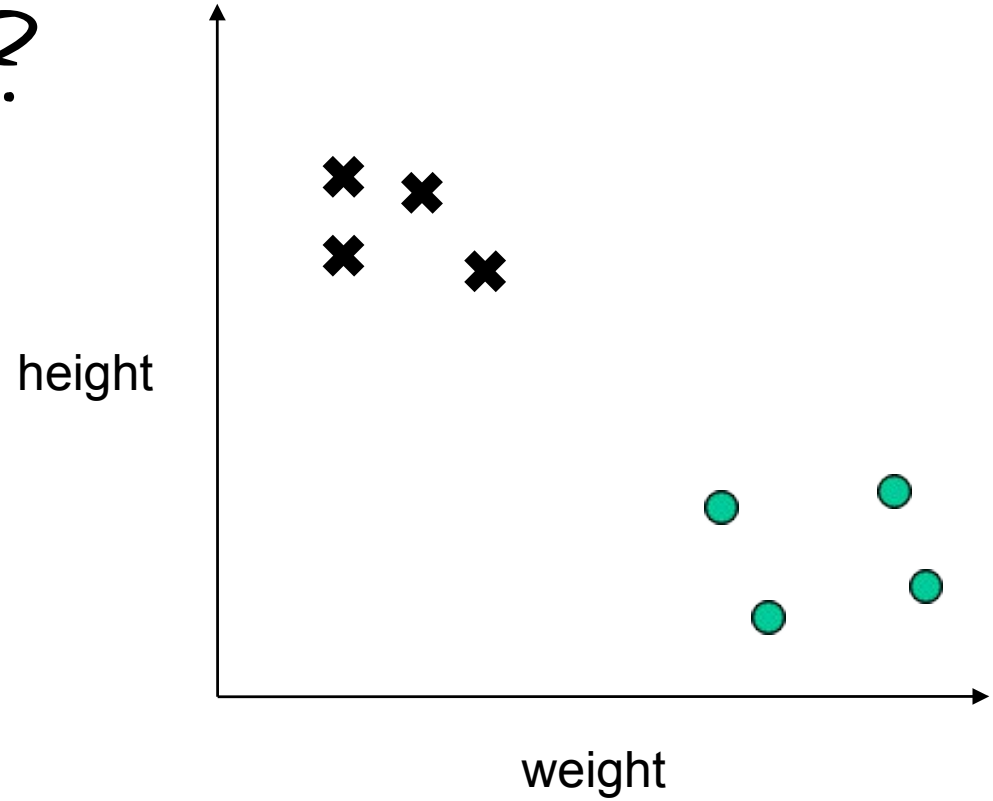
?



Our Problem

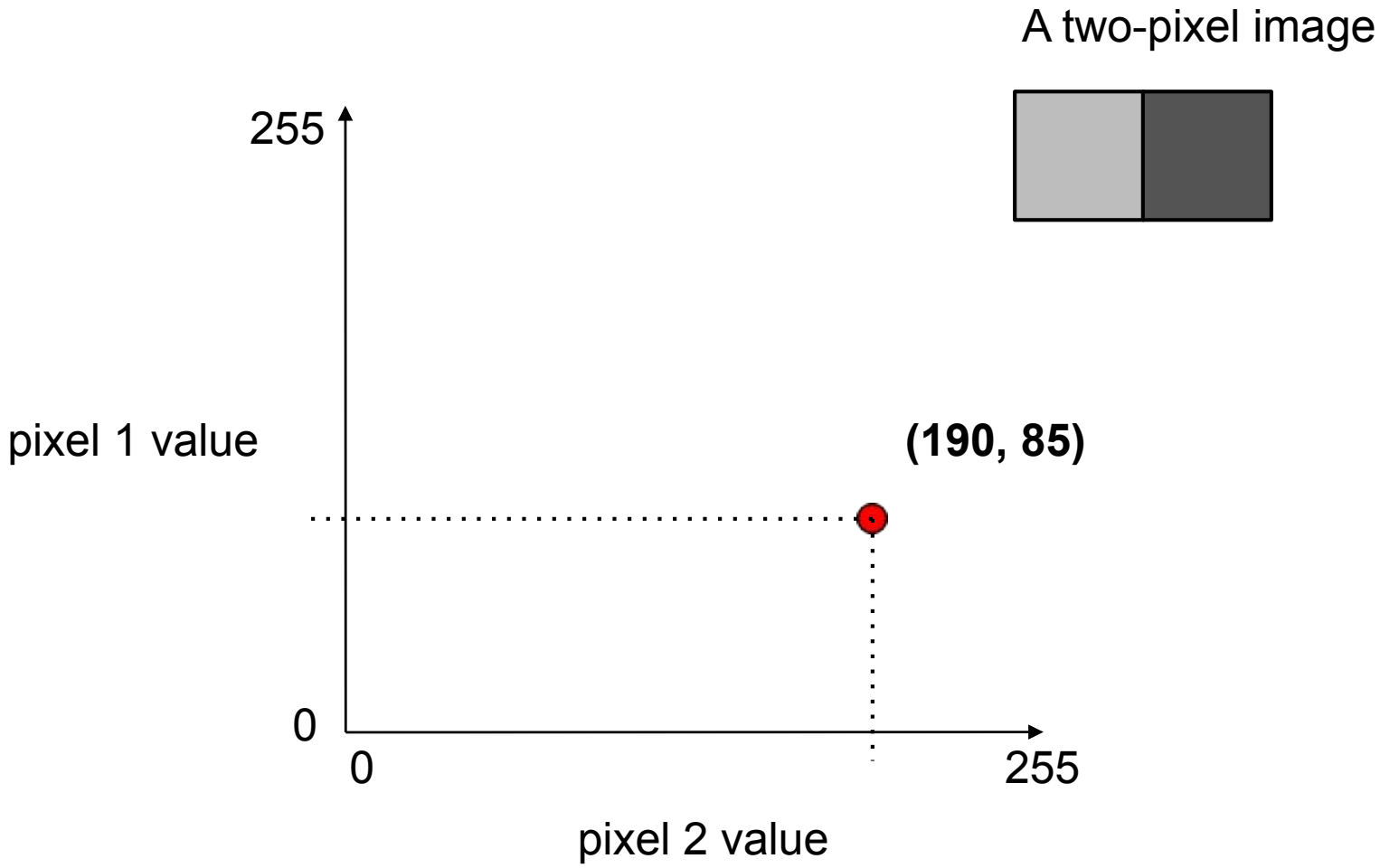
Now – how is this problem like handwriting recognition?



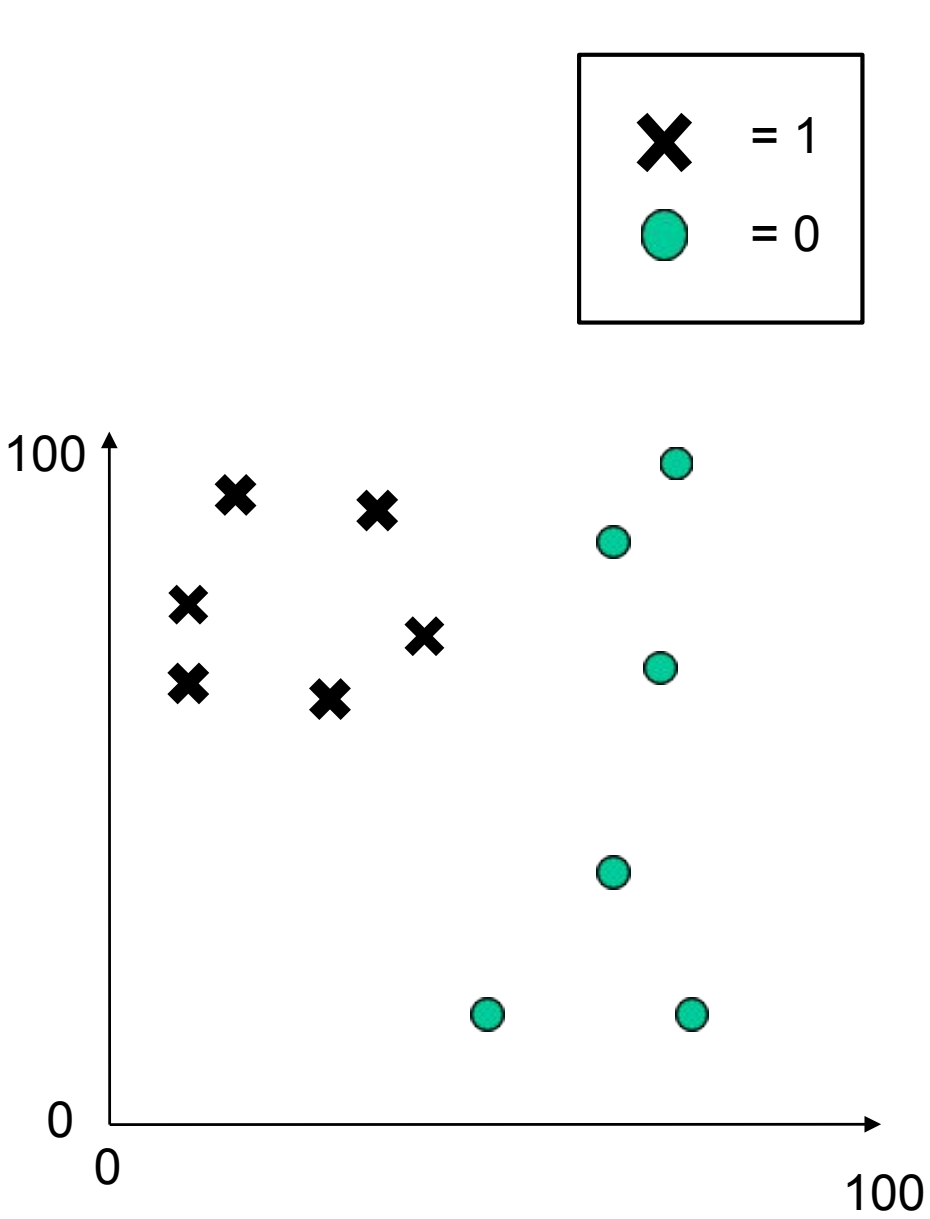


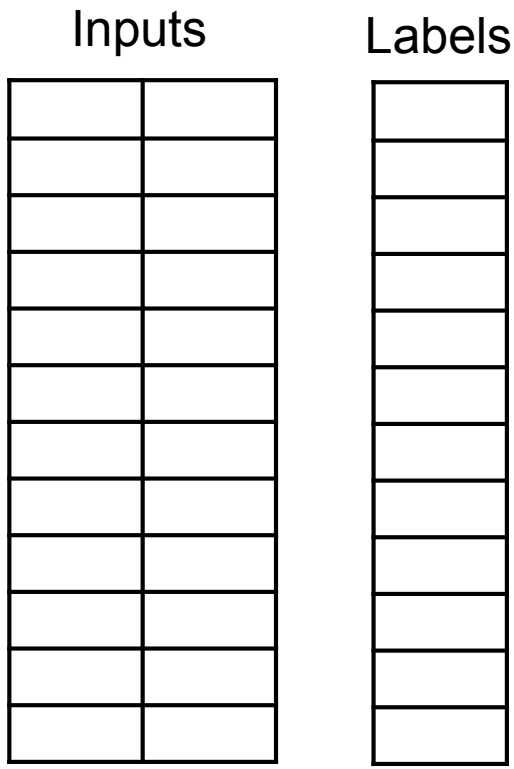
Handwriting Recognition

Let's say the axes now represent **pixel values.**

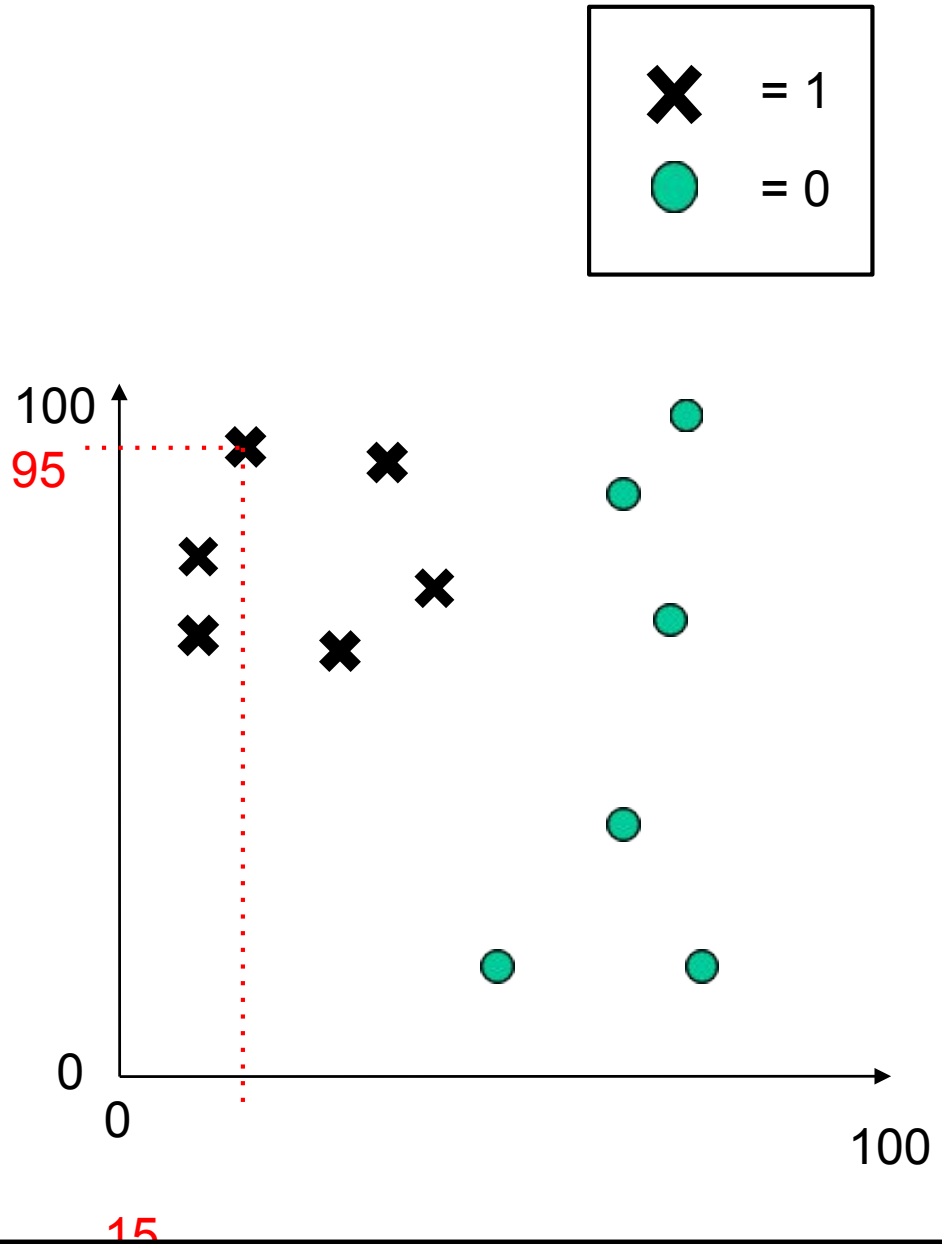


Training and Testing Data





Training and Testing Data



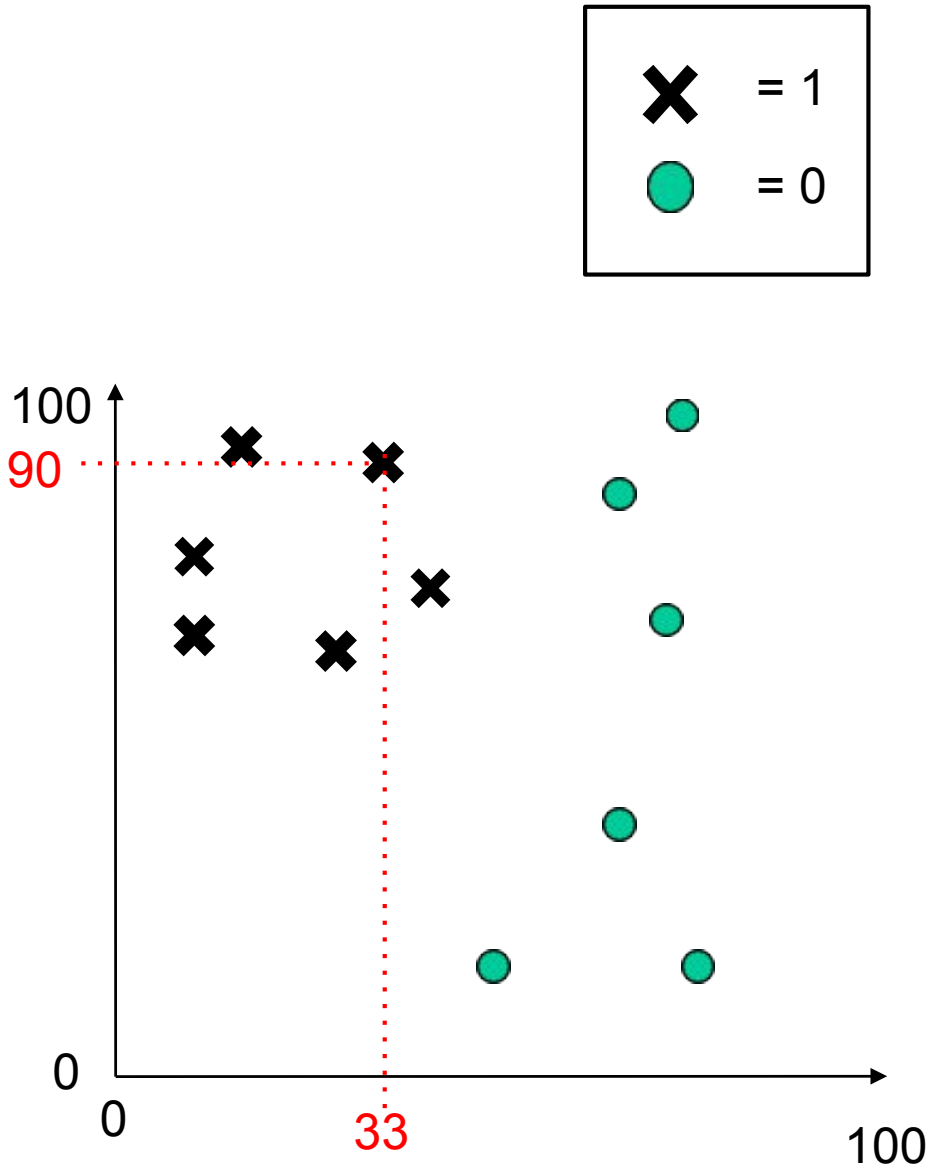
Inputs

15	95

Labels

1

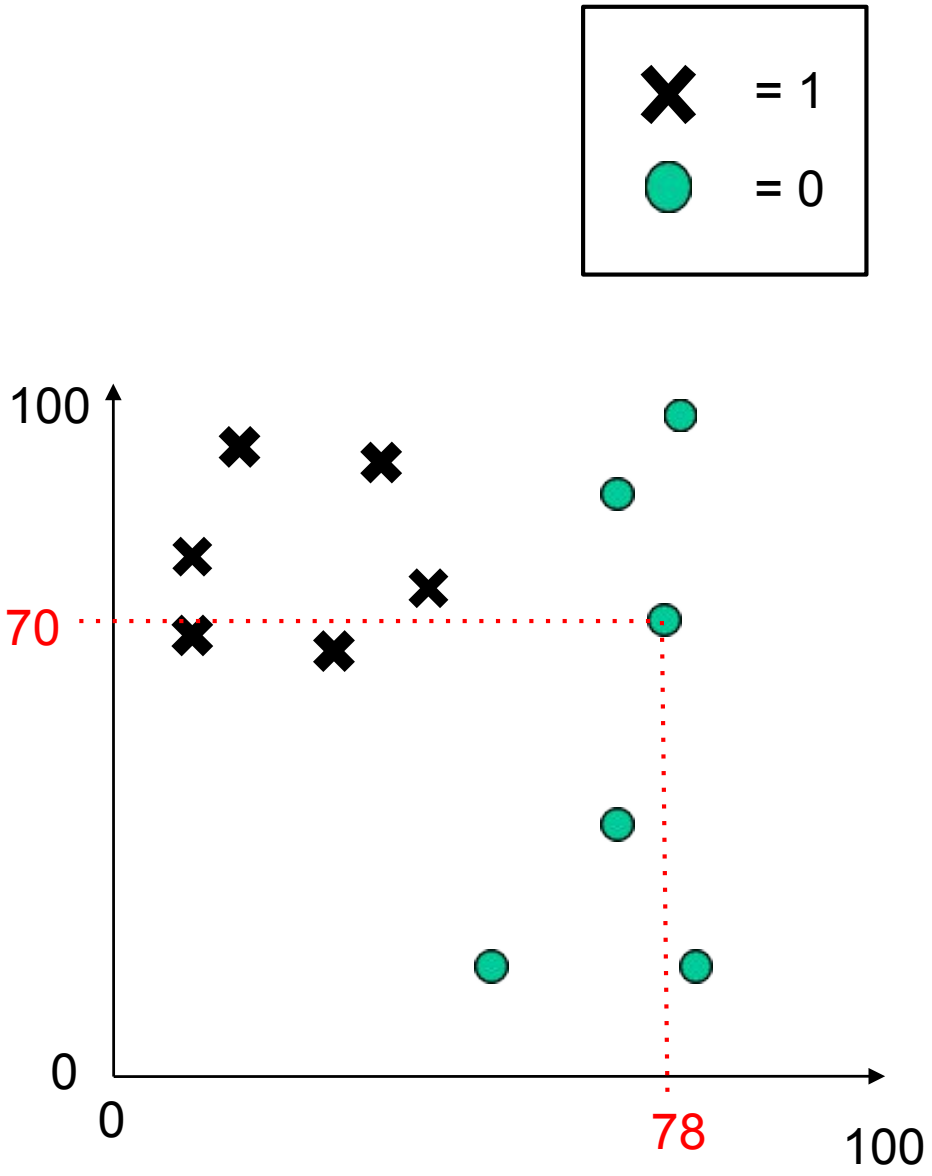
Training and Testing Data



Inputs	
15	95
33	**90**

Labels
1
1

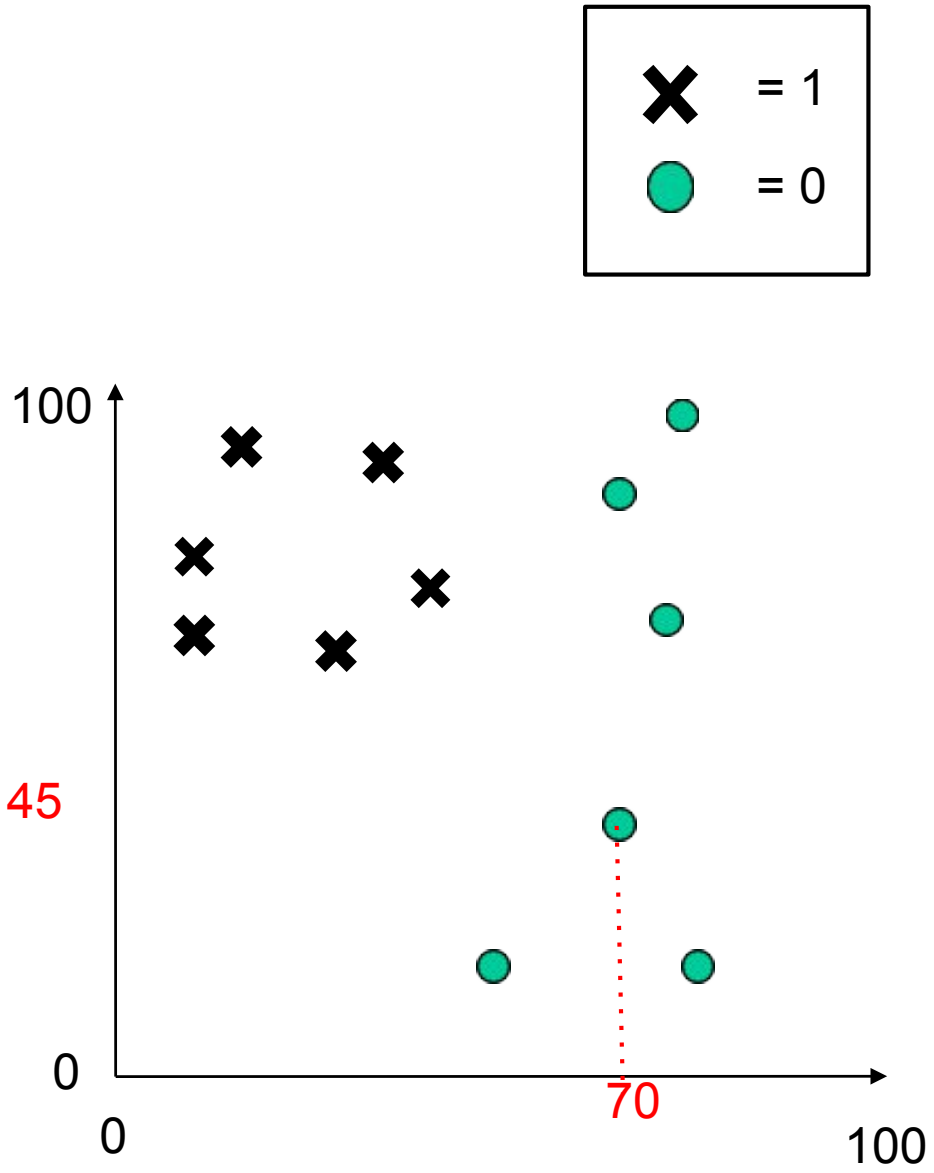
Training and Testing Data



Inputs	
15	95
33	90
78	**70**

Labels
1
1
0

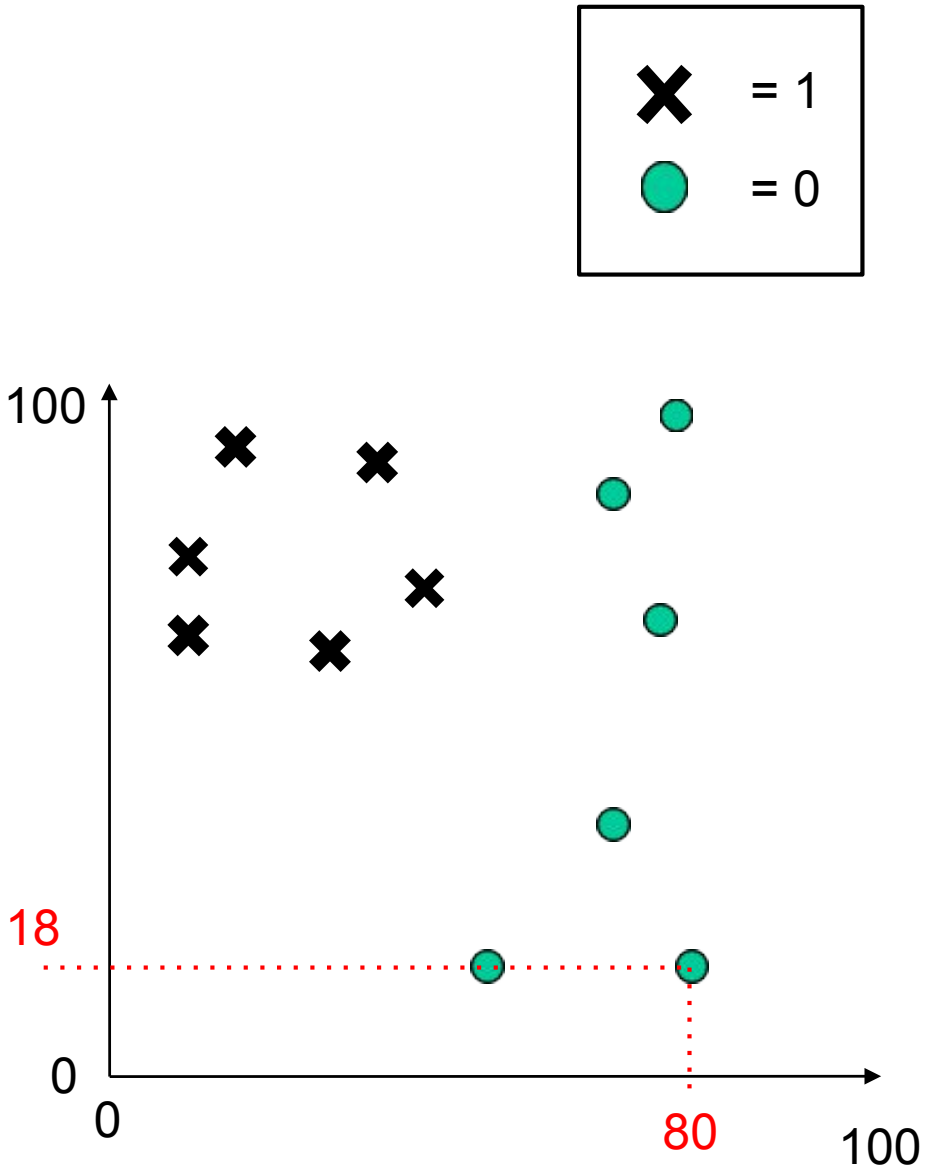
Training and Testing Data



Inputs	
15	95
33	90
78	70
70	**45**

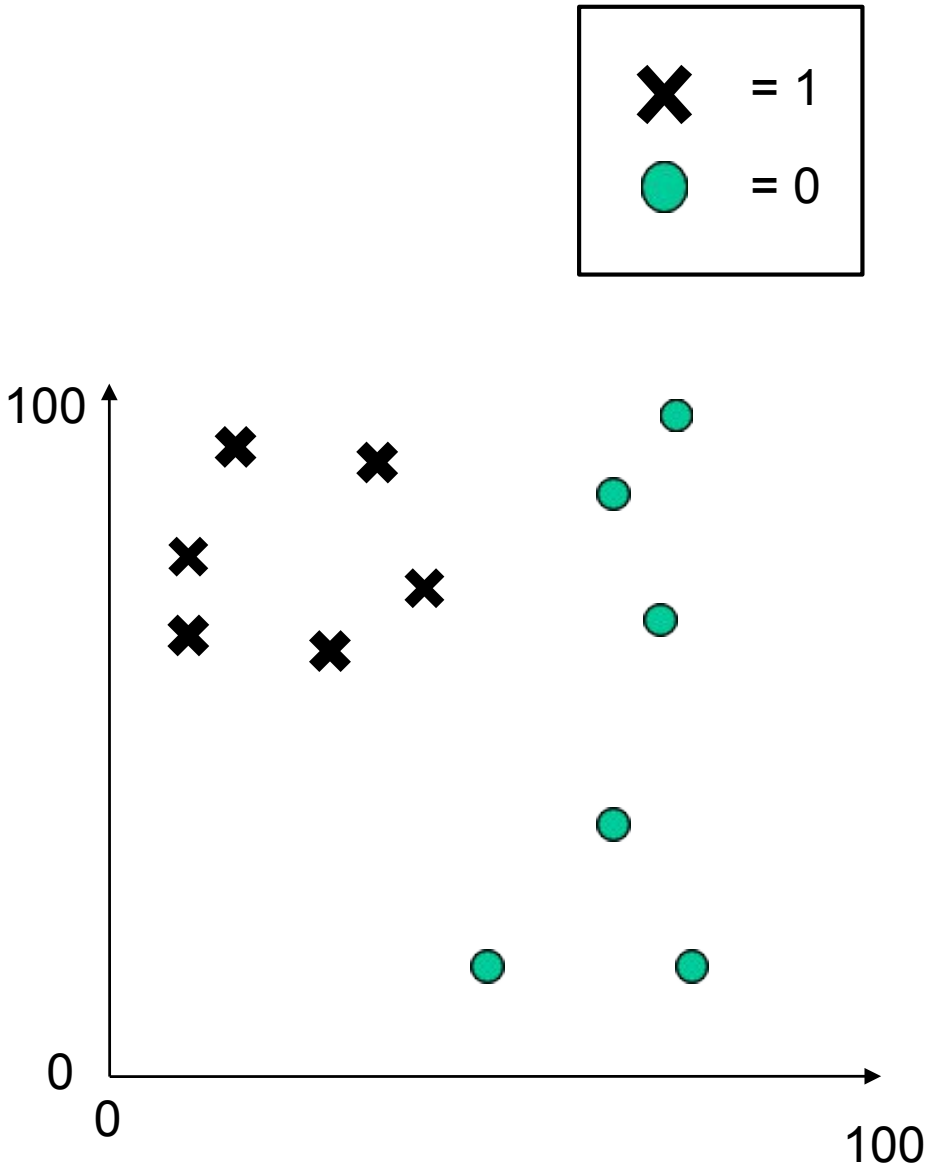
Labels
1
1
0
0

Training and Testing Data



Inputs		Labels
15	95	1
33	90	1
78	70	0
70	45	0
80	**18**	**0**

Training and Testing Data



Inputs		Labels
15	95	1
33	90	1
78	70	0
70	45	0
80	18	0
35	65	1
45	70	1
31	61	1
50	63	1
98	80	0
73	81	0
50	18	0

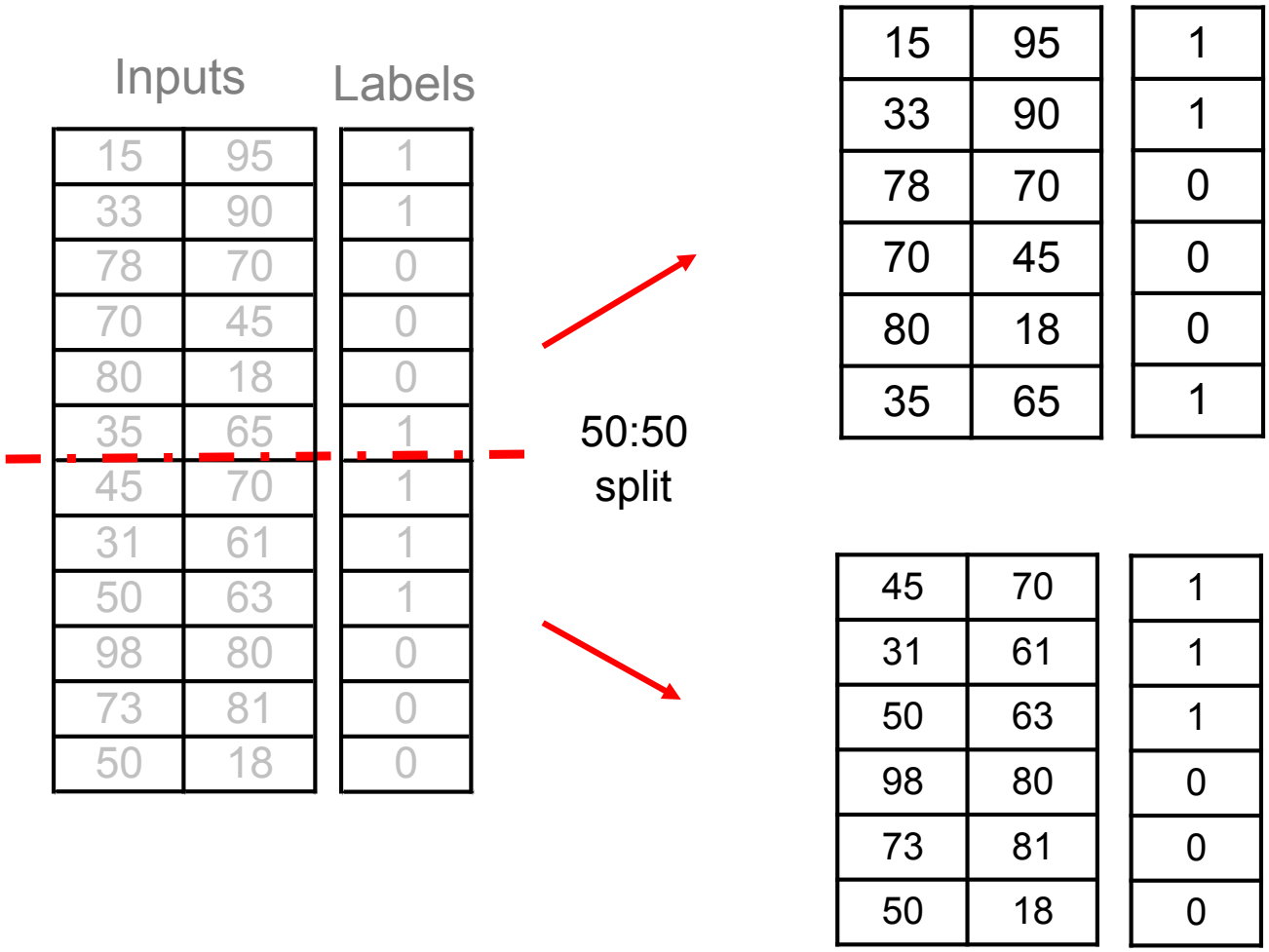
2 “features”
12 “examples”
2 “classes”

Training and Testing Data

Dataset

Inputs		Labels
15	95	1
33	90	1
78	70	0
70	45	0
80	18	0
35	65	1
45	70	1
31	61	1
50	63	1
98	80	0
73	81	0
50	18	0

Training and Testing Data



Training and Testing Data

If too small... you'll make many mistakes on the testing set.

15	95	1
33	90	1
78	70	0
70	45	0
80	18	0
35	65	1

Training set

Train a K-NN on this…

Needs to be big. Bigger is better.

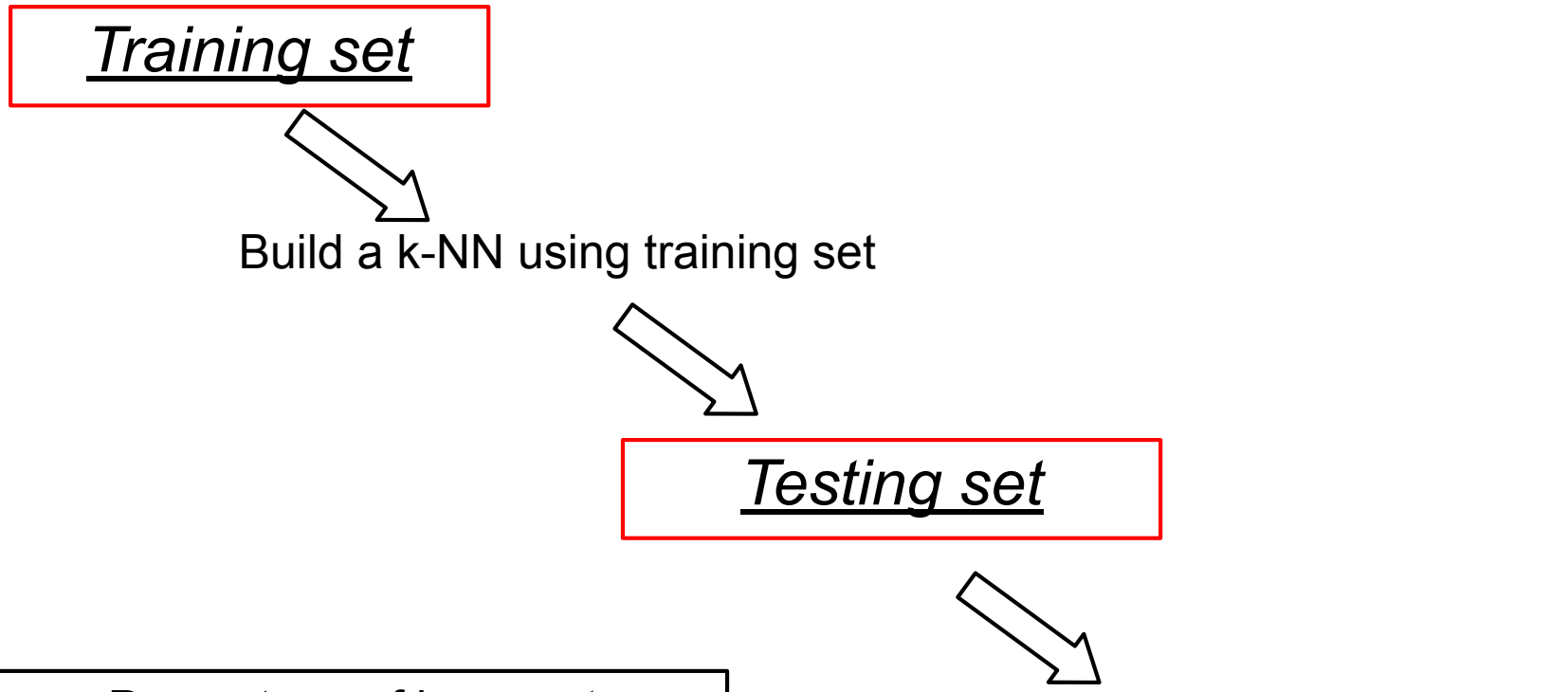
45	70	1
31	61	1
50	63	1
98	80	0
73	81	0
50	18	0

Testing set

… then, test it on this!

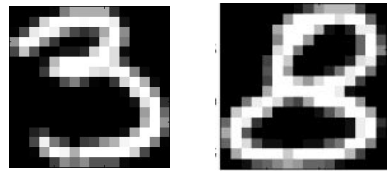
"simulates" what it might be like to see new data in the future

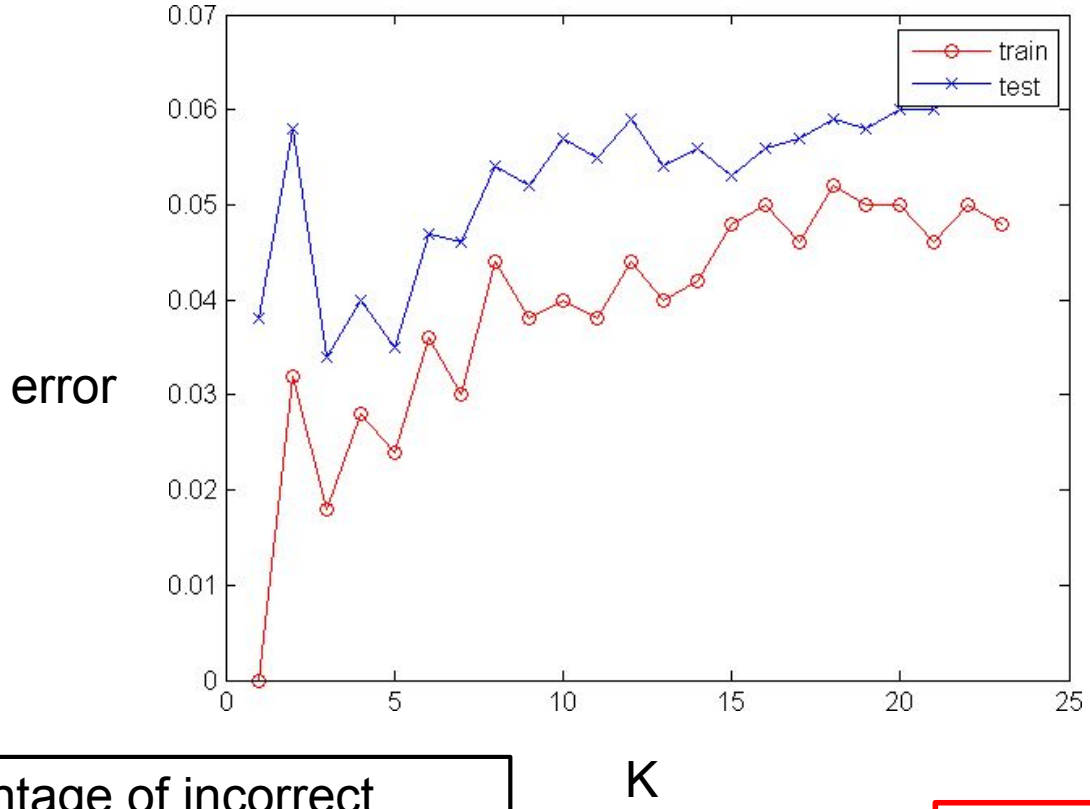
Training and Testing Data



Classifying ‘3’ versus ‘8’ Digits

Plotting error as a function of ‘K’ (as in the K-NN)



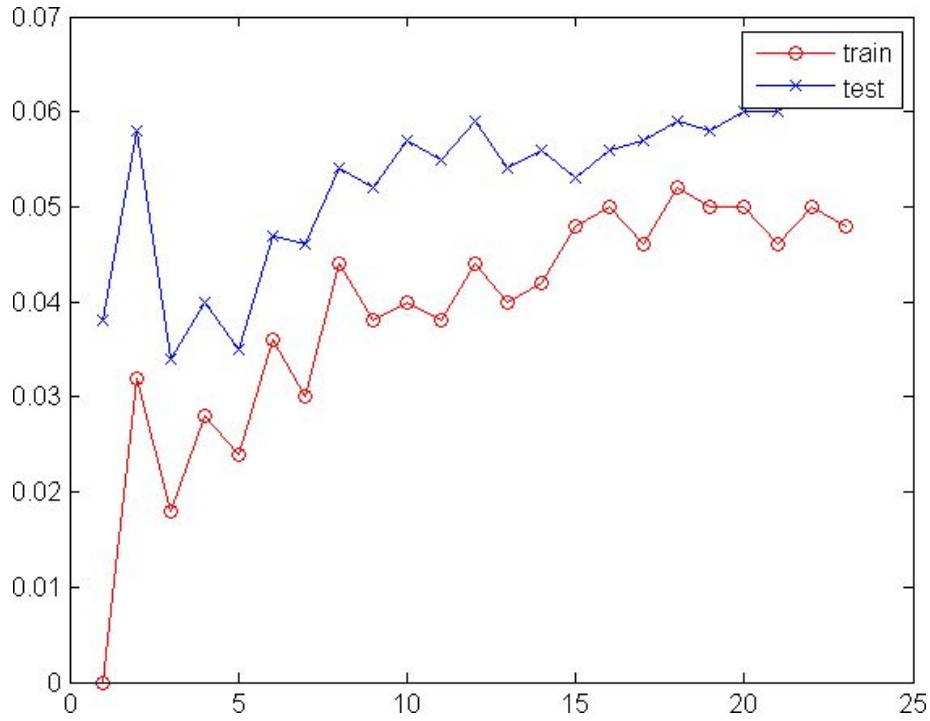


Percentage of incorrect predictions is called the “error”

e.g. “Training” error
e.g. “Testing” error

Training data can behave very differently to testing data!

Classifying ‘3’ versus ‘8’ Digits

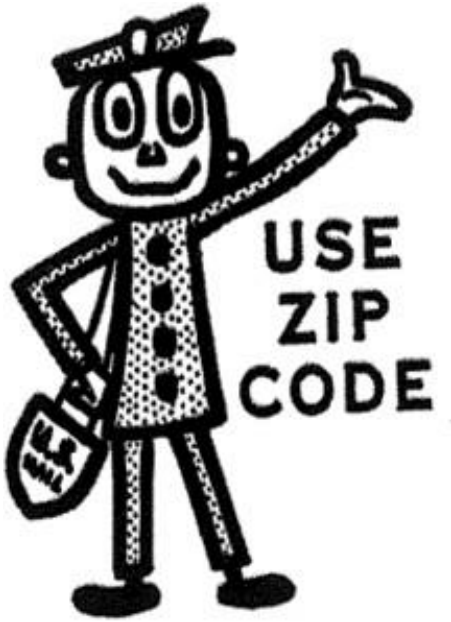


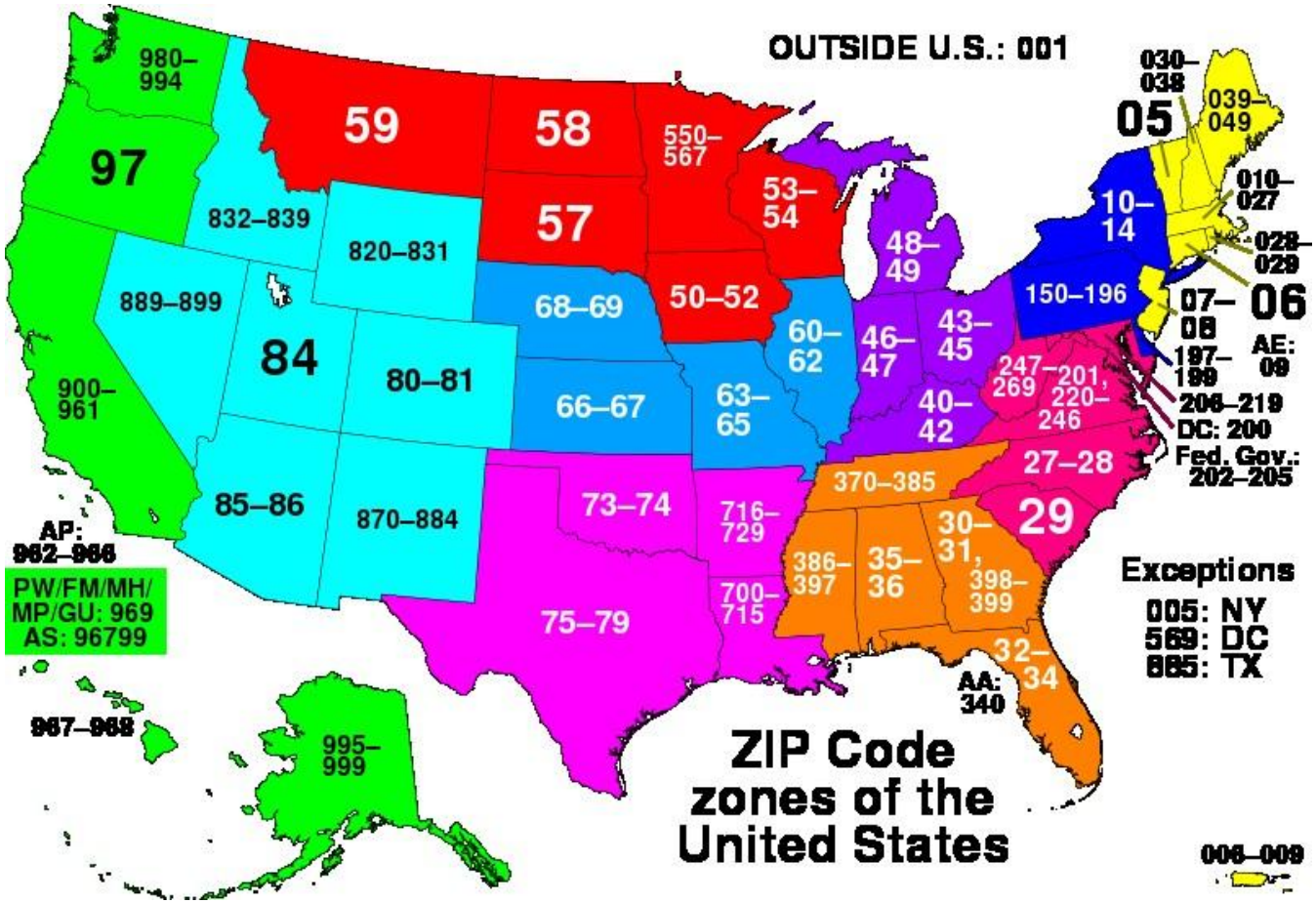
Best testing error at *K*=3, about 3.2%.

Is this “good”?

Classifying ‘3’ versus ‘8’ Digits

Zip-codes: “8” is very common on the West Coast, while “3” is rare. Making a mistake will mean your Florida post ends up in Las Vegas!

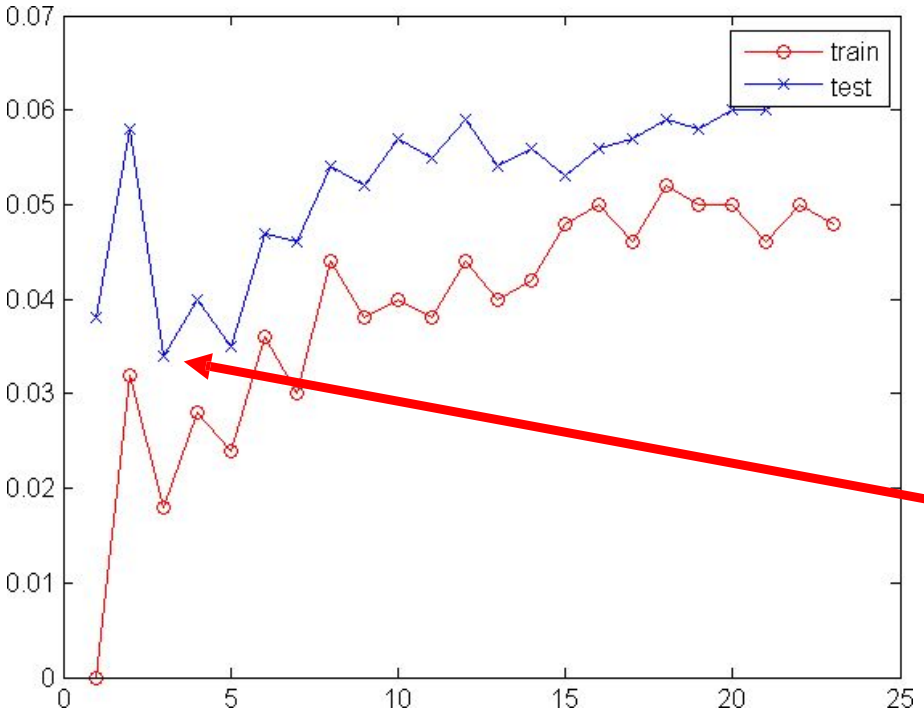


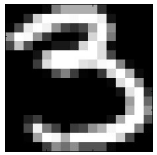


Classifying ‘3’ versus ‘8’ Digits

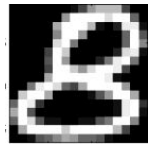
Sometimes, classes are rare, so your learner will not see many of them.

What if, in testing phase you saw 1,000 digits ….





32 instances

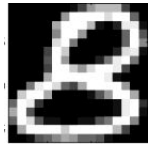


968 instances

3.2% error was
achieved by just saying
“8” to everything!

Classifying ‘3’ versus ‘8’ Digits

Solution?



32 instances

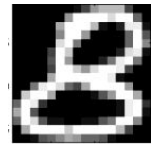
968 instances

0% correct

100% correct

Measure accuracy on each class separately.

R.O.C. Analysis

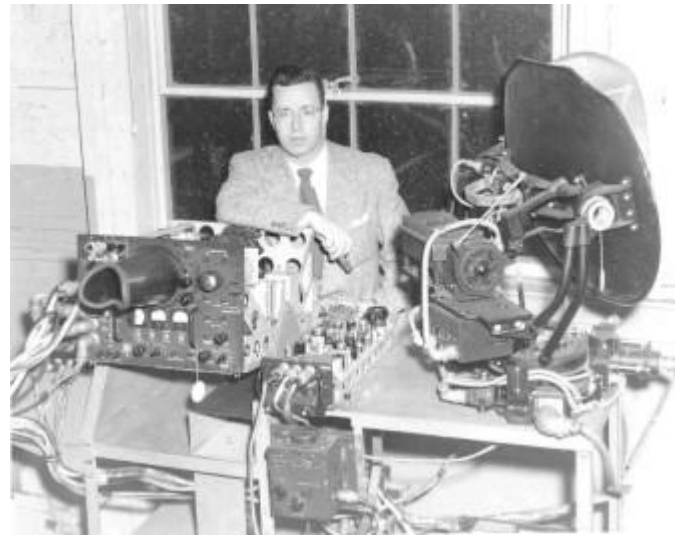


32 instances 968 instances

A statistical framework.

Receiver Operator Characteristics
Developed in WW-2 to assess radar operators.

"How good is the radar operator at spotting incoming bombers?"



False positives
– i.e. falsely predicting a bombing raid

False negatives
–i.e. missing an incoming bomber
(VERY BAD!)

R.O.C. Analysis

The "'3" digits are like the bombers.
Rare events but costly if we misclassify!

False positives – i.e. falsely predicting an event
False negatives – i.e. missing an incoming event

Similarly, we have "true positives" and "true negatives"

		Prediction	
		0	*1*
Truth	*0*	TN	**FP**
	1	**FN**	TP

Building a "Confusion Matrix"

Truth \ Prediction	0	1
0	TN	**FP**
1	**FN**	TP

$$Sensitivity = \frac{TP}{TP+FN}$$

… chances of spotting a "3" when presented with one (i.e. accuracy on class "3")

$$Specificity = \frac{TN}{TN+FP}$$

… chances of spotting an 8 when presented with one (i.e. accuracy on class "8")

R.O.C. Analysis

$$\textit{Sensitivity} = \frac{\text{TP}}{\text{TP+FN}} = ?$$

$$\textit{Specificity} = \frac{\text{TN}}{\text{TN+FP}} = ?$$

Truth \ Prediction	0	1
0	60	**30**
1	**80**	20

TN	**FP**
FN	TP

60+30 = 90 examples in the dataset were class 0

80+20 = 100 examples in the dataset were class 1

90+100 = 190 examples in the data overall